
Real-Time Mobile EEG Hyperscanning: A Precise and Accessible Platform for Social Brain-Computer Interfaces

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Complete List of Authors:	Roque, Thiago; University of Colorado Boulder ATLAS Institute Sun, Ruojia; University of Colorado Boulder Do, Ellen; University of Colorado Boulder ATLAS Institute Llamas, Daniel; University of Colorado Boulder ATLAS Institute Leslie, Grace; University of Colorado Boulder ATLAS Institute
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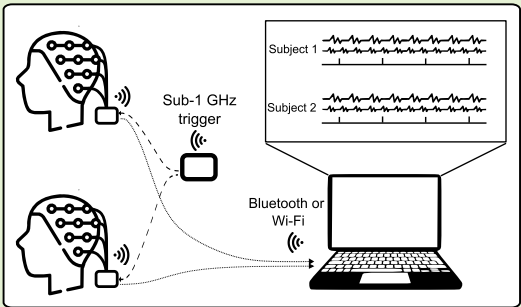
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Real-Time Mobile EEG Hyperscanning: A Precise and Accessible Platform for Social Brain–Computer Interfaces

Thiago Roque, Ruojia Sun, Ellen Do, Daniel Llamas, and Grace Leslie

Abstract—This paper presents a cost-effective fully mobile EEG platform for reliable synchronous multi-brain imaging (hyperscanning) for social human-machine interactions, designed for both online and offline applications. We first quantify the error introduced when Lab Streaming Layer (LSL) is used as primary synchronization method on off-the-shelf mobile EEG systems: sampling-rate drifts and software latencies render conventional wireless hyperscanning techniques unreliable. To overcome these limitations, we introduce a novel wireless hardware-trigger synchronization approach based on sub-1GHz technology with a customized acquisition software, leveraging affordable commercial wireless EEG (OpenBCI Cyton system) to achieve near-zero phase-delay synchronization while maintaining full mobility. The integrated hardware–software stack reduces online inter-device jitter to more than 1/10 of the original observed jitter, yielding a 99.3% of accuracy in phase-locking value and spectral coherence during online measurement of a synthetic 30 Hz signals. A mobile hyperscanning study on interpersonal behavioral synchronization (N = 8) validates ecological performance: significant inter-brain coupling was observed only during well-synchronized steps ($p < 0.05$, FWER-corrected). Finally, we demonstrate real-time visualization of inter-brain coupling using customized version of Vizaj, an open-source spatial networks visualization tool, confirming suitability for live neurofeedback. The proposed synchronization platform costs under \$100 for two heads, requires no cables between participants, and is readily adaptable to OpenBCI hardware. By releasing reproducible designs and code, this work lowers the barrier to group-level neurotechnology and paves the way for social brain-computer interfaces.

Index Terms—brain-computer interface (BCI), hyperscanning, mobile EEG, neurofeedback, sensors synchronization, social interactions, social neuroscience.



I. INTRODUCTION

UNTIL very recently, research on synchronous multi-brain imaging (SMBI) - or hyperscanning - has been limited by expensive, complex, and static brain imaging setups. Despite advances in mobile EEG and the recent emergence of commercial products branded as wireless hyperscanning EEG devices, most of them are costly solutions relying on the Lab Streaming Layer (LSL) protocol for cross device

synchronization, without clear and public specifications on synchronization precision.

By enabling cost-effective, reliable and replicable solutions, SMBI would allow a new paradigm in human-machine interaction, expanding brain-computer interfaces (BCI) from single-user setups to collaborative, multi-brain networks that leverage shared neural data to enhance digital interactions.

Addressing this gap, our work proposes a budget-friendly and fully mobile multi-brain EEG system capable of achieving near-zero phase delay synchronization on real-time applications.

This work proposes the following contributions:

- Systematic evaluation of the Lab Streaming Layer (LSL) protocol for multi-brain EEG synchronization, identifying its limitations for precision hyperscanning.
- Introduction of a novel hardware-software synchronization approach that achieves near-zero phase delay across multiple wireless EEG devices while maintaining full mobility.
- Implementation of the solution using the open-source OpenBCI Cyton platform and a custom wireless synchronization module, ensuring broad compatibility and

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Thiago Roque is with the Brain Music Lab - ATLAS Institute, University of Colorado, Boulder, CO, 80309 USA (e-mail: thiago.roque@colorado.edu)

Ruojia Sun is with the Office for Public and Community-Engaged Scholarship, University of Colorado, Boulder, CO, 80309 USA (e-mail: ruojia.sun@colorado.edu)

Ellen Do is with the ACME Lab - ATLAS Institute, University of Colorado, Boulder, CO, 80309 USA (e-mail: ellen.do@colorado.edu)

Daniel Llamas is with the Brain Music Lab - ATLAS Institute, University of Colorado, Boulder, CO, 80309 USA (e-mail: daniel.llamasmaldonado@colorado.edu)

Grace Leslie is with the Brain Music Lab - ATLAS Institute, University of Colorado, Boulder, CO, 80309 USA (e-mail: grace@colorado.edu)

- scalability for multi-brain research.
- Validation of the proposed system through a real-world mobile hyperscanning experiment on interpersonal coordination, demonstrating precise inter-brain signal alignment during naturalistic social interaction.

Although a specific EEG device was used for development and validation, the system is designed to be broadly compatible, enabling seamless integration with a wide range of EEG platforms.

A. Multi-Brain Brain-Computer Interfaces for Social Interactions

Recent hyperscanning research has identified coupled neural activity across subjects as a potential basis for empathy, mutual understanding and social connection [5], [7], [15], with the degree of inter-brain coupling (IBC) being positively correlated with affective processes and team performance [10]. Such findings reinforce the potential of synchronous multi-brain imaging as tool for enhancing group cohesion and joint actions through multi-brain neurofeedback and social BCI systems.

Early explorations of multi-brain EEG biofeedback can be traced back to the pioneering work of experimental artists such as David Rosenboom and Richard Teitelbaum in the 1960s and 1970s [12]. These initial artistic explorations of passive BCI to modulate human perception opened new possibilities that would later be translated into clinical and therapeutic practices [13].

In contrast to traditional “active” BCI systems, which aim to provide users with direct and voluntary control over the system’s output, “passive” BCIs offer a different paradigm as the system’s input is based on non-intentional brain activity, enabling implicit interactions where the input is inferred rather than actively chosen by the user [17]. These systems present an intriguing model for social BCI, as significant aspects of social interactions occur at an unconscious level, without intentional purpose or voluntary control.

Within the context of hyperscanning and the recent evidence linking inter-brain coupling to social closeness, a handful of recent experiments have attempted to develop real-time neurofeedback systems based on IBC measures to enhance social interactions. Notably, Suzanne Dikker and colleagues have pioneered such effort on a series of artistic and neuroscientific works known as Mutual Wave Projects to explore IBC neurofeedback using commercial mobile EEG headsets [4]. In one of these projects, the Mutual Wave Machine, two participants are enclosed in an intimate capsule and immersed in an audiovisual environment that dynamically responds to their shared brain activity. Greater IBC between the individuals is reflected in more vivid and coherent audiovisual patterns, while a lack of synchronization leads to chaotic sensory feedback. This real-time audiovisual translation allows participants to directly experience and even intuitively manipulate their efforts to connect or distance themselves from their partner. Outcomes from Mutual Wave Projects extend beyond art installations to include studies of real-world dynamic group interactions. Building on the technology developed for

the artistic installations, the same research team adapted the systems for mobile ecological experiments involving students in a classroom [2], [3].

Moving beyond static social interactions, Theofanopoulou et al. (2024) endeavored to apply real-time IBC visualization during a live dance performance. Researchers utilized mobile EEG to simultaneously record brain activity from five experienced dancers while they performed Butoh, a postmodern dance originating in Japan. This interdisciplinary project involved collaboration between dancers, engineers, neuroscientists, musicians, and multimedia artists. The approach focused on examining the inter-brain synchrony among the dancers in relation to auditory cues within the music composed for the performance and investigated changes in synchrony between rehearsals and the final public performance.

While multi-brain BCI systems remain nascent, technical limitations have so far confined their use primarily to artistic explorations rather than practical applications.

II. LSL AS SYNCHRONIZATION METHOD FOR MOBILE MULTI-BRAIN EEGS

The usage of mobile EEG in hyperscanning experiments poses a challenge in assuring precise sample synchronization across devices. Traditional SMBI studies have relied on amplifiers synchronized by a common clock, which require wires (usually optic cables) interconnecting equipment, hindering mobile capabilities. Recent studies using commercial mobile EEG devices have relied mostly on the LSL protocol to synchronize the multiple data streams. Despite the fact that this method permits the usage of a high number of synchronized devices, with considerable simplified and affordable setups, LSL synchronization capabilities are heavily dependent on hardware and networking infrastructure.

LSL is an open-source communication protocol designed specifically for synchronizing multimodal physiological data streams. It employs a peer-to-peer time synchronization algorithm modeled after the Network Time Protocol (NTP) where each incoming sample receives a timestamp from a local monotonic clock. This architecture is capable of decomposing timing errors into constant offsets, slow drifts, and high frequency jitters.

Although optimized implementations on a single computer can achieve multi-stream synchronization at $\approx 1ms$ precision, a typical LSL communication presents latencies of $\approx 50ms$ [8]. While this level of precision might be well suited for most behavioral and cognitive experiments requiring synchronization between stimuli and neurological responses, it may prove insufficient for precise inter-brain signal alignment, critical to most inter-brain coupling metrics.

On top of that, LSL is only capable of synchronizing data samples after they are delivered to the LSL protocol. Any hardware or software delay preceding the acquisition of the samples by LSL cannot be accounted without specific optimizations and software modification [9]. One common issue on using independent wireless acquisition systems is the natural variability of sampling clocks within its nominal sampling rate. Taking the example of the ADS1299 ADC

(Texas Instruments), one of the most commonly used analog front end (AFE) in commercial mobile EEGs, its internal clock presents an accuracy of $\pm 0.5\%$ at 25°C . In the case of a sampling rate of 250 Hz (commonly observed in commercial mobile EEGs), two “identical” devices could present effective sampling rates between 248.75 Hz and 251.25 Hz. Even disregarding differences in streaming lags, such variations in sampling rates can easily disrupt inter-brain coupling measures due to phase delays and drifts.

One way to leverage LSL synchronization is through synchronization triggers at hardware level. By using triggers sampled at the same time basis of the physiological data acquisition, we can ensure a shared common time marker across devices.

Based on this solution, researchers have recently proposed a custom-made wireless EEG device with a shared clock for the analog-to-digital converter through a wired RJ45 connection across devices, ensuring sub-sample synchronization at hardware level. Despite solving the synchronization issue, this solution requires a wired connection across amplifiers, limiting its mobile capabilities [1].

III. EVALUATING OPENBCI FRAMEWORK FOR SMBI WITH LSL

The OpenBCI Cyton+Daisy system is a wireless, open-source EEG platform capable of recording up to 16 channels using either dry or wet electrodes. While the Cyton board can support sampling rates up to 250 Hz, the Daisy expansion module limits its sampling rate to 125 Hz when used with its default Bluetooth radio frequency (RF) module. The open-source nature of the hardware and software allows for customization, including access to three external analog and five digital data inputs, which can be sampled synchronously with physiological signals.

To evaluate the feasibility of the OpenBCI platform for real-time mobile hyperscanning, we assessed three Cyton+Daisy devices using OpenBCI’s native software interface (GUI v5.2.1) with Lab Streaming Layer (LSL) integration. Our evaluation targeted two key factors critical for synchronized multi-brain imaging: sampling jitter and the variability of both sampling rate and latency.

In our experimental setup, we delivered continuous, synchronous wired-connected triggers at three-second intervals to pairs of Cyton devices. Each device was connected to a dedicated instance of the OpenBCI GUI on a same computer, with each GUI streaming data to LSL via its networking widget. The resulting LSL streams were captured using LabRecorder. For each Cyton pair, we conducted four independent trials, each consisting of a 5-minute recording session. To simulate realistic mobile recording conditions, all devices were placed three meters from the Bluetooth receiver dongle.

In a second analysis, we assessed the impact of phase delay - resulted from differences in sampling rates - by calculating spectral coherence (SC) and phase locking value (PLV) over two synthetic sinusoidal signal of high beta brain frequency (28 Hz), sampled at the average effective sampling rate plus $\pm 1\text{ SD}$, obtained from the previous experiment. PLV and SC were calculated from (1) and (2), respectively.

$$PLV_{xy} = \left| \frac{1}{N} \sum_{n=1}^N e^{-i(\theta_x(n) - \theta_y(n))} \right| \quad (1)$$

$$SC_{xy}(f) = \frac{|X(f)Y(f)^*|^2}{(X(f)X(f)^*)(Y(f)Y(f)^*)} \quad (2)$$

A. Results

We observed a significant variability on the sampling rate of the EEG signal by the Cyton+Daisy device. Despite the nominal sampling rate of 125 Hz, the average effective sampling rate observed in the LSL recordings was 125.116 Hz ($SD = 0.151$ Hz).

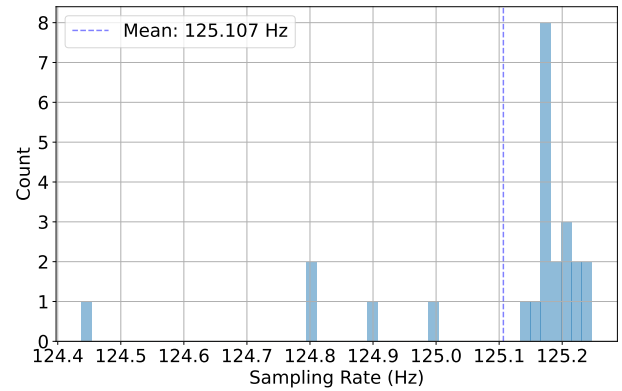


Fig. 1. Effective sampling rates observed by LSL using the Cyton+Daisy

If we take, for example, two sinusoidal signals of 20 Hz, sampled at 124.965 Hz and 125.267 Hz ($125.116 \pm 1\text{ SD}$) and with perfect phase alignment at $t = 0$, in ≈ 4 seconds they will present phase delay of 90° .

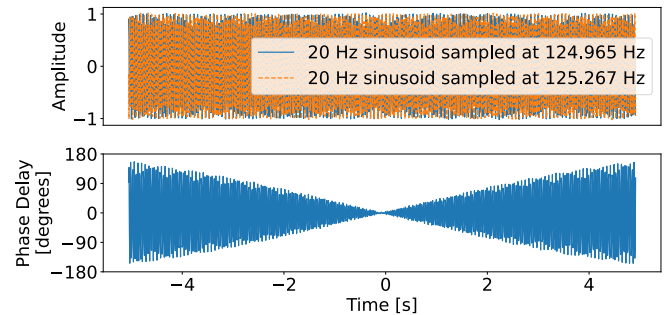


Fig. 2. Example of phase delay on the observed sampling rate variation

From these results, we synthesized two sinusoidal signals of 28 Hz, representing high beta frequencies, sampled at 124.965 Hz and 125.267 Hz and phase aligned at $t = 0$, and calculated SC and PLV for varying window lengths centered at $t = 0$. For windows larger than 3.4s, both connectivity measures presented errors larger than 10% due to the presence of phase delay.

A second issue observed with the OpenBCI GUI was inconsistency in the sampling rate of the triggers. Although the EEG amplifier samples auxiliary inputs at the same nominal rate as the EEG signal (125 Hz), the effective sampling rate

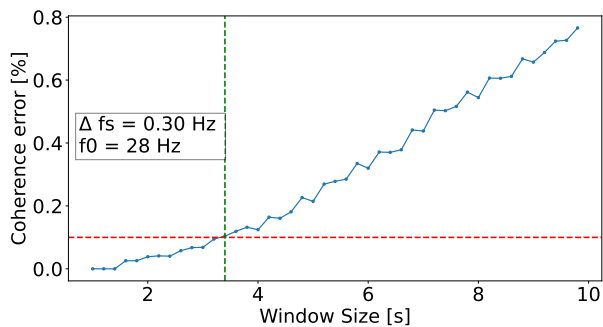


Fig. 3. Coherence error (%) vs. Window Size on the presence of phase delay across signals

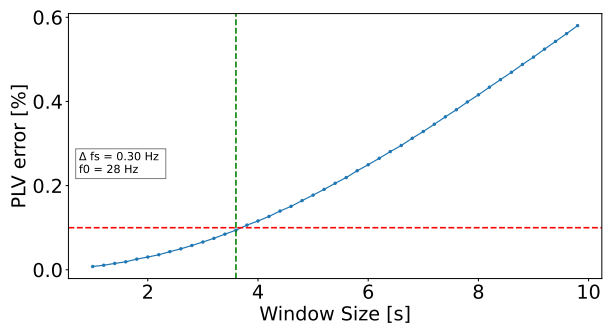


Fig. 4. PLV error (%) vs. Window Size on the presence of phase delay

observed in LSL was approximately 70 Hz (mean = 70.21 Hz, $SD = 1.78$ Hz). This discrepancy appears to arise from software limitations in the OpenBCI GUI's LSL integration, resulting in high variability in trigger intervals, increased jitter, and inconsistent delays. Such inconsistency can be noted through the high standard deviation observed in measurements of inter-trigger intervals (mean = 3.0566s, $SD = 412$ ms).

The average inter-device trigger latency measured was 500ms ($SD = 864$ ms, median = 244ms), which is substantially higher than the precision required for accurate hyper-scanning studies. These findings indicate that, in its current configuration using the OpenBCI GUI and LSL, the OpenBCI Cyton+Daisy system does not provide the temporal accuracy necessary for synchronized multi-brain imaging applications.

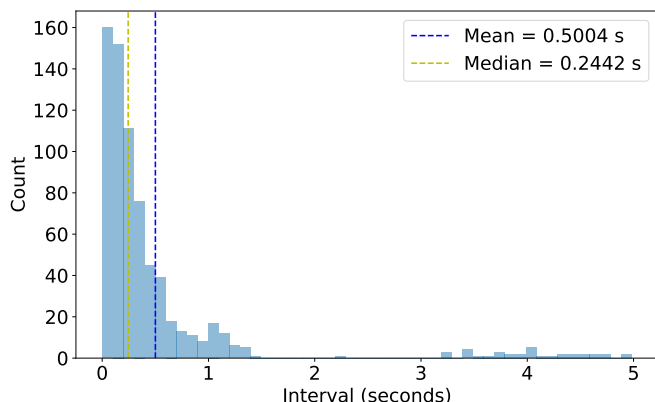


Fig. 5. Histogram of the inter-device trigger latency using the OpenBCI native framework

While the OpenBCI hardware is flexible and cost-effective, the presented high variability underscores significant software overheads that cannot be compensated by the LSL protocol — particularly in the GUI's network streaming implementation — limiting its effectiveness for applications requiring precise temporal synchronization.

IV. CUSTOM OPTIMIZED SOFTWARE AND WIRELESS TRIGGER FOR MULTI-STREAM SYNCHRONIZATION

To improve stability and the required time synchronization across devices during signal acquisition, we developed a custom software based on the BrainFlow framework allowing for multi-threading data acquisition from the Cyton board and linkage to LSL. In contrast to the method employed by the OpenBCI GUI, in which EEG data and auxiliary (trigger) data results in different LSL streams, we opted to generate a single multi-channel stream composed of all EEG channels plus all triggers, distinguishable by stream type. This method ensures that EEG data and trigger events are aligned and associated with a common timestamp, facilitating processing for both real-time and offline applications.

Through this new optimized software, acquisition delays were reduced to an average of 23ms ($SD = 10$ ms). For offline applications, pre-processing steps based on LSL timestamp matching and upsampling can improve acquisition delays to subsample level (≈ 1 ms).

To achieve near-zero phase delay in real-time applications, while maintaining complete wire independence, we developed a wireless synchronization trigger based on the CC1310 sub-1GHz transceiver. We chose the CC1310 Sub-1GHz device due to its very low current consumption, ideal to reduce interference in the EEG voltage supply, and capability to easily implement custom protocols for minimum transmission delays. Considering a custom protocol with a symbol rate of 400 Kbit/s, Rx filter bandwidth of 622 kHz, preamble of 2 Bytes, and payload of 4 bytes (including address), transmission delay is expected to be less than 200 μ s with a packet error rate under 0.5% withing a range of 10 meters and Tx power of 14dBm.

We designed a custom PCB to be attached to the Cyton board capable of forwarding up to 4 independent triggers to Cyton's auxiliary inputs. The complete system adopts a controller/target architecture, consisting of one sub-1GHz receiver connected to each Cyton EEG device, a primary transmitter serving as the controller unit, and a central computer that collects all data streams. A custom Python software aggregates all synchronized data streams and forwards them to LSL. The controller unit can be configured to send triggers both continuously, at regular intervals, and at specific moments of interest, based on the experimental paradigm, at different auxiliary inputs. For this paper we employed the LaunchXL-CC1310 development board, from Texas Instruments, as our controller unit.

Focusing on real-time applications, we tested the system using a synthetic 30 Hz sinusoidal signal with a 1 mV amplitude and additive white noise. This signal was simultaneously delivered to two Cyton+Daisy biosignal acquisition boards, each equipped with the wireless trigger receiver module. The



Fig. 6. Sub-1GHz synchronization trigger module attached to a Cyton board

controller device was configured to transmit synchronization pulses every three seconds. To assess the system's temporal precision, we first evaluated the consistency of trigger reception by measuring inter-device trigger latency and analyzing the variance in trigger intervals across both devices.

Subsequently, we quantified synchronization performance by calculating SC and PLV of the synthetic signals acquired by the synchronized boards. These metrics were computed using two windowing strategies: event-locked windows, defined as 6 s epochs centered on each trigger, and continuous windows of fixed 6 s duration that disregarded trigger timing. Acquired signals by the Cyton+Daisy boards were captured by our custom software, forwarded to LSL, and recorded using LabRecorder. Offline analysis was then conducted in MNE-Python.

A. Results

Regarding sampling rate variation, we observed no significant difference between our system and the OpenBCI GUI, except by the fact that triggers are now captured and streamed long with the EEG, resulting in a single stream with multiple data types following a same timestamp.

Across all three devices tested, the average inter-device trigger latency was 42.1ms ($SD = 90.5ms$), with a median value of 12.7ms. A more complete understanding of the inter-device latency can be seen in the histogram presented by Fig. 7.

It's worth noting the presence of a low, but significant number of observed latencies above 70 ms. Throughout the experiment we noted an increasing variability on timing interval of the triggers as the batteries of the Cyton devices started to get low. Out of 15 recording sessions, 3 were removed due to inconsistent data because of the aforementioned reason. Despite the fact that the CC1310 module is a low-power device, we hypothesize that the extra current load during trigger reception moments resulted in voltage fluctuations that interfered in the Bluetooth packet transmission.

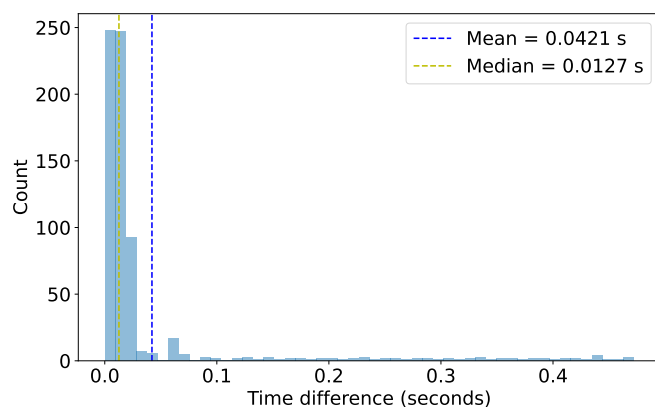


Fig. 7. Histogram of the inter-device trigger latency using the sub-1GHz module

The average inter-trigger interval, transmitted by the wireless modules and captured by the custom software, was 3.0607 s ($SD = 3.51ms$).

Results from the connectivity measures of the sinusoidal signal sent simultaneously to two Cyton devices can be seen in Figs. 8 and 9. The average PLV and SC obtained was 0.9925 and 0.9920, respectively.

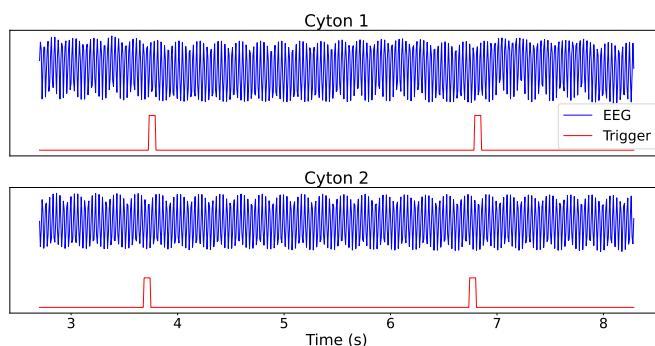


Fig. 8. Example of the synthetic signal sent to both boards with triggers received through the wireless module

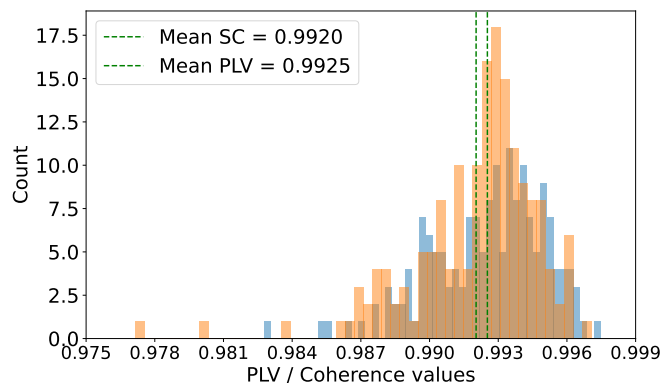


Fig. 9. Histogram of the calculated SC and PLV using the wireless trigger and a synthetic signal. Epochs of 6 s centered at the triggers

V. ANALYSIS OF THE RESULTS

Compared to data acquisition using the OpenBCI GUI, our proposed Python-based software demonstrated significant improvements in both acquisition latency and timing stability. To quantitatively assess these differences, we conducted a Wilcoxon rank sum test, which revealed that the Python-based software achieved inter-device trigger latencies significantly better than those observed with the OpenBCI GUI, with an improvement of approximately $450ms$ ($W = 448380$, $z = 26.37$, $p < 0.0001$, $r = 0.703$). This result indicates a substantial reduction in latency.

Although the mean inter-trigger intervals were similar between the two systems ($3.06s$ for the Python software vs. $3.05s$ for the OpenBCI GUI), the precision of timing differed significantly. A Levene's test confirmed that the variance in inter-trigger intervals was dramatically lower with the custom software ($p < 0.0001$), with measured variances of $0.0034s$ and $0.4205s$ for the Python software and OpenBCI GUI, respectively. These findings highlight the superior timing stability and precision achieved by our optimized acquisition framework.

Despite some residual inter-device delay in the custom software—primarily due to inherent variability in effective sampling rates—this issue can be effectively addressed by epoching the data streams around trigger events. By aligning epochs in this manner, we enable the calculation of connectivity measures from signals acquired at virtually identical time points, achieving sub-sample timing precision across devices.

VI. EXPERIMENTAL VALIDATION OF THE SYTEM

This study was approved by the Institutional Review Board of the University of Colorado, Boulder, (Protocol 23-0596). All participants provided informed consent prior to participation, in accordance with the Declaration of Helsinki.

To validate the potential of the OpenBCI Cyton+Daisy EEG in a real mobile interaction we designed an hyperscanning study on interpersonal behavioral synchronization on tango dancers.

In this study, participants were instructed to perform synchronized forward and backward dance steps while assuming fixed roles of leader or follower, according to the traditional tango practice. Throughout the experiment, EEG and inertial measurement unit (IMU) data for each ankle of the dancers were recorded. We used ankle movements as behavior markers of good or bad interpersonal coordination moments.

5 couples were recruited for this experiment (4 male-female and 1 female-nonbinary pairs), with participants having at least three years of tango experience ($\mu = 12$ years, $SD = 7.8$). Due to insufficient artifact-free trials, data from one couple had to be excluded, resulting in 4 couples used during the analysis process.

We recorded four trials of 3 minutes per couple. Prior to the task, participants were asked to stand still for one minute to record baseline EEG data, which was used for motion artifact cleaning and establishing a baseline brain connectivity measure, without social interaction.

Using a 19-channel passive Ag/AgCL sintered wet cap connected to the Cyton+Daisy EEG, we recorded brain activity

for each pair using 16 electrodes following the standard 10-20 international system (Fp1, Fp2, F3, F4, T7, C3, Cz, C4, T8, P7, P3, Pz, P4, P8, O1, O2), with CPz as reference and AFz as ground. Electrode impedances were maintained below $100k\Omega$, in accordance with manufacturer's guidelines. Behavioral data were captured using four 6-degrees-of-freedom (6-DoF) IMU modules (MPU-6050), connected to an ESP32, positioned on each participant's ankles to detect foot movement instances. Data was transmitted via Wi-Fi using the Open Sound Control (OSC) protocol to the same computer used for EEG acquisition, which was appended to LSL as a separate stream.

Instants of foot movements were identified through an amplitude thresholding of the principal component analysis decomposition (3 components decomposition) of the 6 axis IMU data, followed by a wavelet filtering (coif3 wavelet, 3 levels decomposition) and a low pass filtering at 2 Hz. Detected instants of movement (IoM) for each foot were appended to the EEG data as markers. To compensate for desynchronization due to software latencies and divergences in effective sampling rate, we up-sampled all streams to 500 Hz, and corrected sample lagging by comparing timestamp.

Based on the detected IoM, we defined synchronized and non-synchronized behaviors as:

- Synchronized behavior: Instants of movement (IoM) between the leader and follower with temporal distances smaller than 200 milliseconds.
- Non-synchronized behavior: IoM with temporal distances between 600 milliseconds and 1 second.

We preprocessed EEG data using MNE-Python. Initial FIR filtering (linear phase, Hamming window) limited the signal bandwidth to $0.5 - 45Hz$. Bad channels, identified through visual inspection, were removed, interpolated, and later discarded from brain connectivity analysis. Artifact subspace reconstruction (ASR) was applied to remove motion artifacts using the pre-task 1 minute of clean EEG and a cutoff of $15 SD$. Post-ASR, EEG data was re-referenced to a common average (CAR), and independent component analysis (ICA) was used to remove residual eye artifacts.

We segmented the clean EEG data into epochs from $-1.5s$ to $0.5s$ relative to the behavioral markers. Epoch length was chosen to minimize interference from consecutive steps. We calculated IBC for all possible electrode combinations (excluding bad electrodes) within each dyad using SC, for theta ($4 - 7 Hz$), alpha-mu ($8 - 12 Hz$), and beta ($14 - 30 Hz$) frequency bands.

Statistical significance was assessed by cluster-based permutation test, using dependent samples t-test as statistic of interest, comparing the measured IBC from the dyad with a surrogate data composed of an epoch shuffling across each dyad. This shuffling technique ensured statistical similarity of the surrogate dataset while removing connectivity from social interactions. Cluster threshold was defined through threshold-free cluster enhancement (TFCE).

Validating the system, we found statistically significant IBC only during good interpersonal synchronization, but none for bad interpersonal synchronization.

For beta frequency bands, we observed significant IBC between right pre-frontal regions of the leader to central

regions of the follower ($t(270) = 2.828$, $p < 0.021$ (FWER corrected), $d = 0.283$).

For alpha-mu frequency bands, we observed significant IBC on the synchronized behavior condition mainly from pre-frontal and right regions of leaders' brains to central and right central regions of followers' brains ($t(270) > 2.387$, $p < 0.041$ (FWER corrected), $d = 0.180$).

For theta frequency bands, we observed significant IBC between centro-parietal regions (Pz, P3, and P4) of leaders to left parietal regions of followers ($t(270) > 2.782$, $p < 0.044$ (FWER corrected), $d = 0.200$).

For the non-synchronized behavior conditions, we observed no significant IBC in either frequency band ($p > 0.1$).

Our findings are consistent with previous hyperscanning studies, which have shown that increased IBC is positively correlated with effective interpersonal coordination. These results confirm that the OpenBCI Cyton EEG system, when used with an optimized synchronization framework, is a viable and reliable platform for mobile hyperscanning research.

VII. REAL-TIME HYPERSCANNING FEEDBACK

To validate the real-time capability of the system, we designed a proof-of-concept real-time visualization for IBC based on a modified version of the Vizaj software [11]. Vizaj is an open-source software designed for interactive visualization of spatial networks. The original code was modified to permit the reception of a live data stream coming from a custom Python algorithm using websockets. Nodes were also customized to reflect proximal positions of real EEG electrodes on a scalp for the given dataset.

The real-time workflow of this system is represented in Fig. 10. The Sub-1 GHz transmitter was programmed to emit synchronization triggers every 3 seconds, which is received by all EEG devices and is appended to the LSL streams synchronized to the EEG data. A real-time LSL receiver software was designed to parse all LSL streams and epoch them in four second windows centered at the trigger markers. This allows for a continuous stream of epoched data while assuring precise synchronization across devices with minimal errors. If an epoch presents EEG levels higher than a predetermined threshold (set to 200uV), that epoch is classified as bad and discarded.

EEG artifacts are attenuated by a combination of Artifact Subspace Reconstruction (ASR) and Spatial Harmonic Analysis (SPHARA), in real-time. To enable more natural interactions, involving more movements, we chose a more aggressive cutoff for ASR (8 SD). To smooth transitions, IBC measures from the last five epochs are averaged and compared to a baseline IBC calculated from the resting dataset (recorded during no interaction). Any averaged IBC higher than the baseline is sent to Vizaj and plotted for real-time visualization.

Using recorded data from the dance experiment, we could replicate and validate the implementation of the real-time visualization of IBC by playing back the EEG data synchronously to the video recordings of different experimental sessions. The real-time processing was implemented by epoching EEG data with fixed 6s windows, with an overlap of 3s. Using an Intel

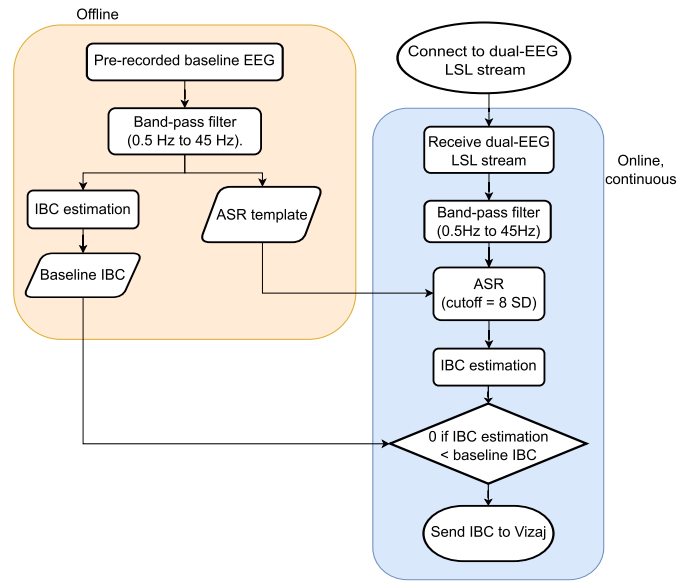


Fig. 10. Flowchart of the designed real-time system for IBC visualization

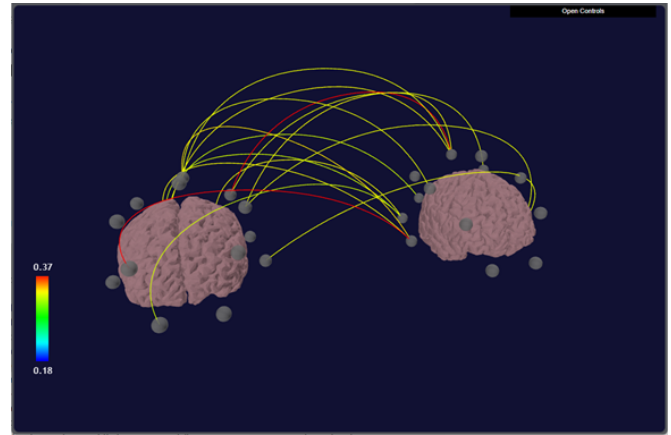


Fig. 11. Example of the IBC visualization designed on Vizaj

Core i7-10750H CPU @2.60 GHz, 32 GB RAM memory, the average execution time of the algorithm is 400ms, accounting for the real-time FIR filtering (band-pass from 1Hz to 45Hz), artifact subspace reconstruction for artifact reduction, IBC estimation using the Coherence measure, baseline correction from pre-session recordings, websocket communication from the Python script to Vizaj, and the Vizaj rendering of the significant inter-brain connections.

VIII. CONCLUSION

We have presented a novel, cost-effective, and fully mobile multi-brain EEG system designed to achieve near-zero phase-delay synchronization in real-time, addressing a critical limitation in the field of social brain-computer interfaces. By systematically evaluating the performance of the widely adopted Lab Streaming Layer (LSL) with OpenBCI Cyton+Daisy devices, we demonstrated that software-based synchronization alone is insufficient for SMBI applications requiring precise temporal alignment, due to inherent sampling variability and software-induced delays.

To overcome these constraints, we developed an optimized acquisition framework that integrates a wireless sub-1GHz synchronization trigger and custom software, enabling the collection of timestamped, multi-device EEG and trigger data within a unified stream. Our validation with synthetic signals confirmed a substantial improvement in synchronization accuracy, with inter-device trigger latencies reduced to near the sampling interval and connectivity metrics (PLV, SC) consistently above 0.99.

The proposed solution has an overall B.O.M. cost of \$20.00 USD per target EEG plus \$30 USD for the controller. If employed with two OpenBCI Cyton+Daisy EEGs along with two 19-channels sintered electrode cap, the system allows for a complete fully mobile hyperscanning setup under \$7100.00 USD, less than 1/7 of a standard commercial solution.

Experimental validation in a naturalistic hyperscanning study involving tango dancers further demonstrated the system's capacity to capture meaningful inter-brain coupling during moments of behavioral synchrony. Statistically significant IBC was observed exclusively during well-coordinated movement, particularly in alpha-mu and beta frequency bands, highlighting the system's suitability for studying social coordination and the neural mechanisms underlying interpersonal synchrony.

Finally, we implemented a proof-of-concept real-time feedback and visualization pipeline, confirming the feasibility of live IBC monitoring and neurofeedback using the proposed platform. This advancement opens new avenues for applied research in group neuroscience, social neurofeedback, and interactive human-machine systems, enabling robust, scalable, and accessible multi-brain studies beyond the laboratory. In future iterations we plan to incorporate other methods to improve artifact mitigation in real-time. The combination of ASR and SPHARA presented satisfactory results regarding motion artifacts and muscle artifacts, but a substantial quantity of ocular artifacts was still present. As an alternative, by including head motion, neck electromyography, and electrooculography data, we might be able to achieve better EEG cleaning results with adaptive noise cancelling H_∞ filters [6].

Our results establish a foundation for the next generation of social BCI and hyperscanning applications, with the potential to transform research on group dynamics, collaborative performance, and digital systems that adapt to collective neural states, while maintaining a focus on accessibility, mobility, and replicability in real-world contexts.

APPENDIX I LINK TO REPOSITORIES

Project files and gerbers for the Sub-1GHz trigger module can be downloaded at:

https://github.com/Brain-Music-Lab/OpenBCI-Wireless_Trigger

Python script and configuration files for the custom acquisition software can be downloaded at:

<https://github.com/Brain-Music-Lab/HyperOpenBCI-LSL>

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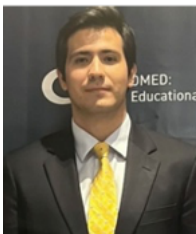
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Thiago R. Roque was born in São Bernardo do Campo, SP, Brazil, in 1988. He holds an M.Sc. in Electrical and Computer Engineering and a BS. in Electrical Engineering with a minor in music technology from the University of Campinas, Campinas, SP, Brazil.

Since 2021 he has been a Fulbright grantee pursuing a triple major Ph.D. degree at the University of Colorado Boulder in Creative Technology, Neuroscience, and Cognitive Sciences. Currently he is a Graduate Research Assistant at the Brain Music Lab working with multi-brain computer interfaces for creative interactions. His research interests include mobile brain/body imaging, brain-computer interfaces, biofeedback, and multi-brain EEG. Along with his academic career, Thiago has over 10 years of experience in the technology industry in R&D, innovation and project management.



Daniel Llamas-Maldonado was born in November 22nd, 1995 in Morelia, Michoacán, México. He received his bachelor degree in electronics engineering at the Universidad Michoacana de San Nicolás de Hidalgo in 2019. He received his Masters in Science degree in electronics engineering at the Tecnológico Nacional de México at Morelia. He is currently coursing a triple track Ph.D. program at the University of Colorado Boulder in Creative Technology, Neuroscience, and Cognitive Sciences. His research is mainly focused on instrumentation and digital signal processing.



Ruojia Sun has over 5 years of experience in human-computer interaction research, dedicated to designing technology to support people to live healthier, more creative lives. Ruojia's research, spanning textile-based wearable devices and technology for dance and movement, has won Best Paper and Best Demo awards at the Designing Interactive Systems conference. She holds an M.S. in Creative Technology & Design from the University of Colorado Boulder, and an M.Eng. and B.S. in Mechanical Engineering from Cornell University. Currently, she is the Grants and Data coordinator at CU Boulder's Office for Public and Community-Engaged Scholarship, where she bridges her research background, technical skills, and passion for social impact by advancing engaged scholarship work across the university.

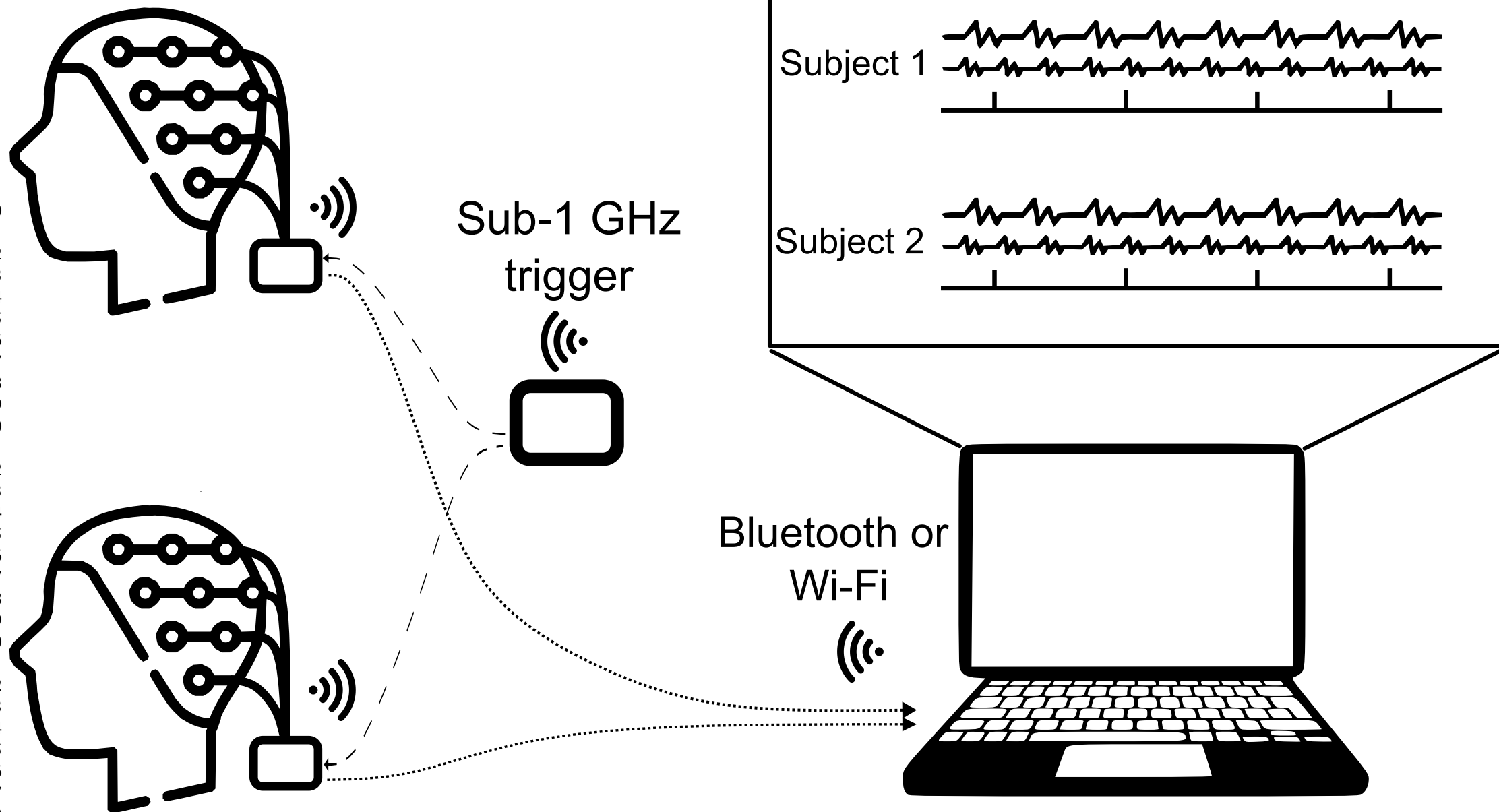


Grace Leslie received the B.A. and M.A. degrees in Music, Science and Technology from Stanford University, Stanford, CA, USA, and the Ph.D. degree in Music and Cognitive Science from the University of California, San Diego, CA, USA, where she conducted research at the Swartz Center for Computational Neuroscience. She was a SUTD-MIT Postdoctoral Fellow at the MIT Media Lab, Cambridge, MA, USA, and later a Neukom Fellow at Dartmouth College, Hanover, NH, USA, before joining the faculty of the Georgia Institute of Technology, Atlanta, GA. Since 2022 she has been an Assistant Professor with the ATLAS Institute and College of Music, University of Colorado Boulder, where she directs the Brain Music Lab. Her research interests include brain- and body-computer interfaces for expressive music performance, sensorimotor synchronization, and music-based biofeedback for health and well-being.

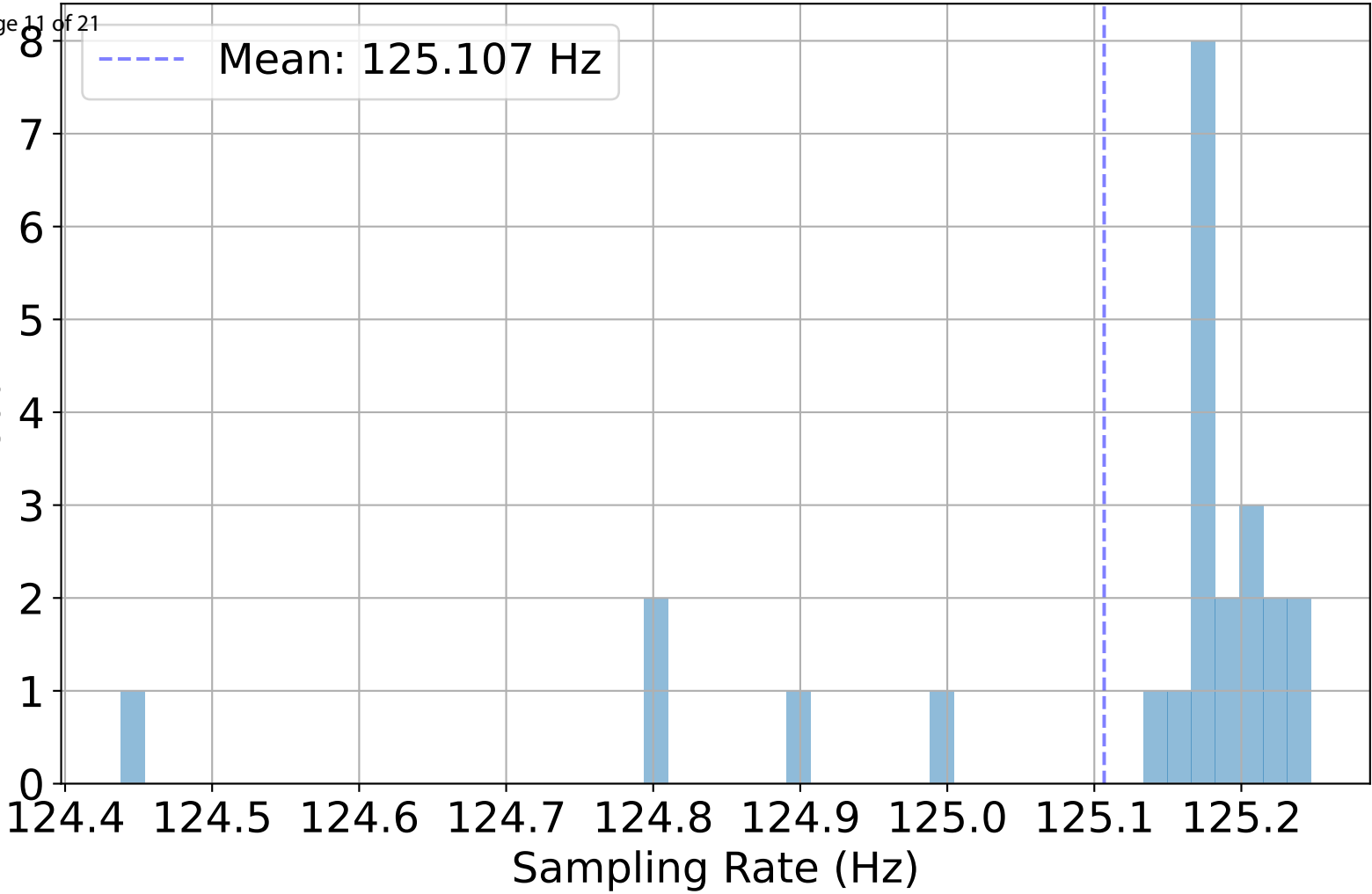


Ellen Yi-Luen Do holds a Ph.D. in Design Computing from Georgia Institute of Technology, a Master of Design Studies from the Harvard Graduate School of Design, and a bachelor's degree from National Cheng Kung University in Taiwan. She has served on the faculties of the University of Washington, Carnegie Mellon University, and Georgia Institute of Technology.

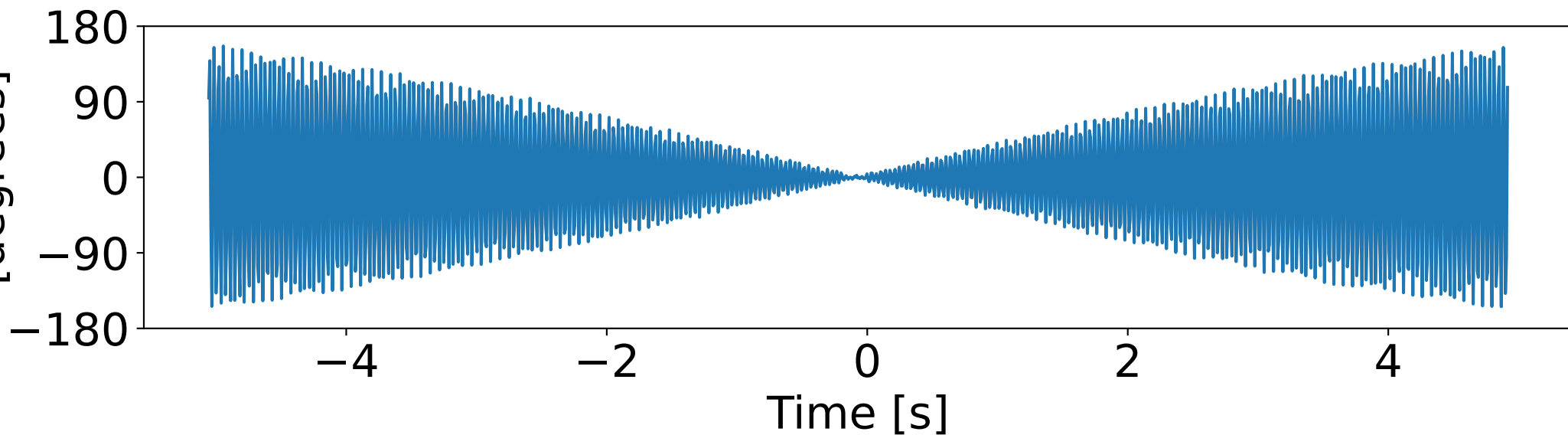
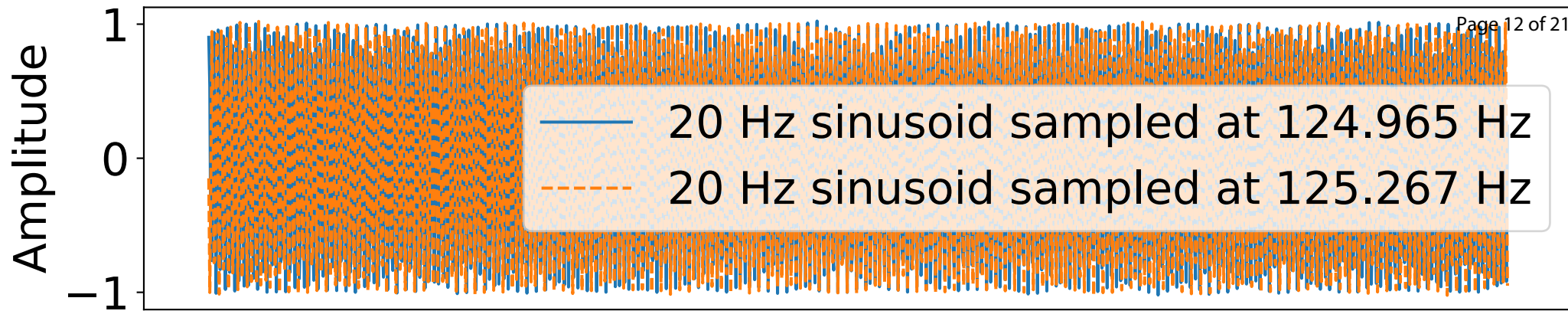
From 2013 to 2016, she co-directed the Keio-NUS CUTE Center in Singapore, a research unit investigating Connected Ubiquitous Technology for Embodiments. Ellen has published over 170 papers and raised over \$11M in research funding. She was Design Community Chair for ACM CHI (2012 and 2013), and currently serves on the Steering Committees for ACM Conference series of TEI (Tangible, Embedded and Embodied Interaction, since 2011) and C&C (Creativity & Cognition, since 2017), ISWC (International Symposium on Wearable Computers, since 2018), and AH (Augmented Human, since 2015), and on the editorial board for AIEDAM (Artificial Intelligence for Engineering Design, Analysis and Manufacturing), Springer Series on Cultural Computing, and AUSMT. She enjoys playing the Djembe and performs at festivals and other venues with the Sensua Players.

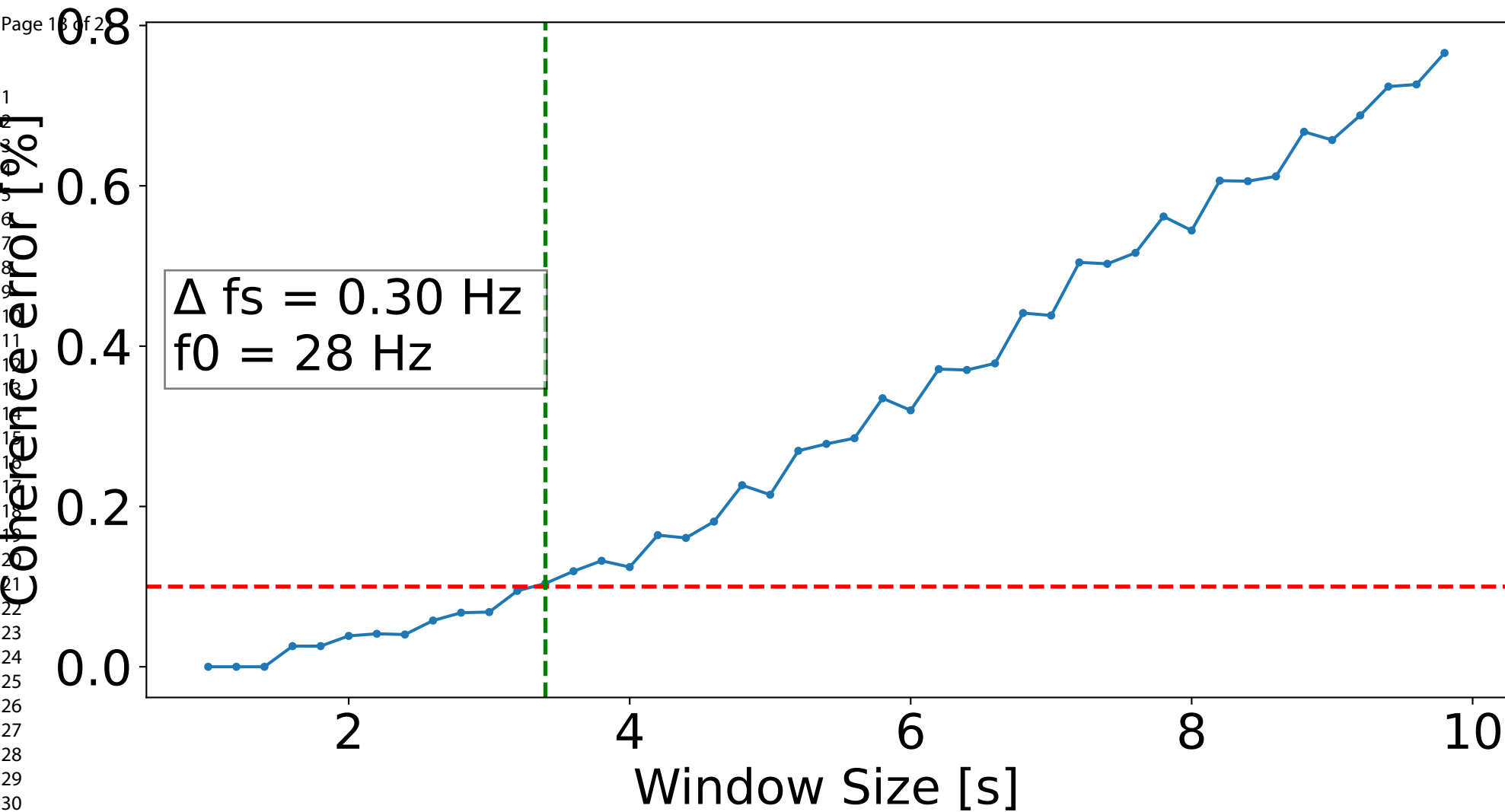


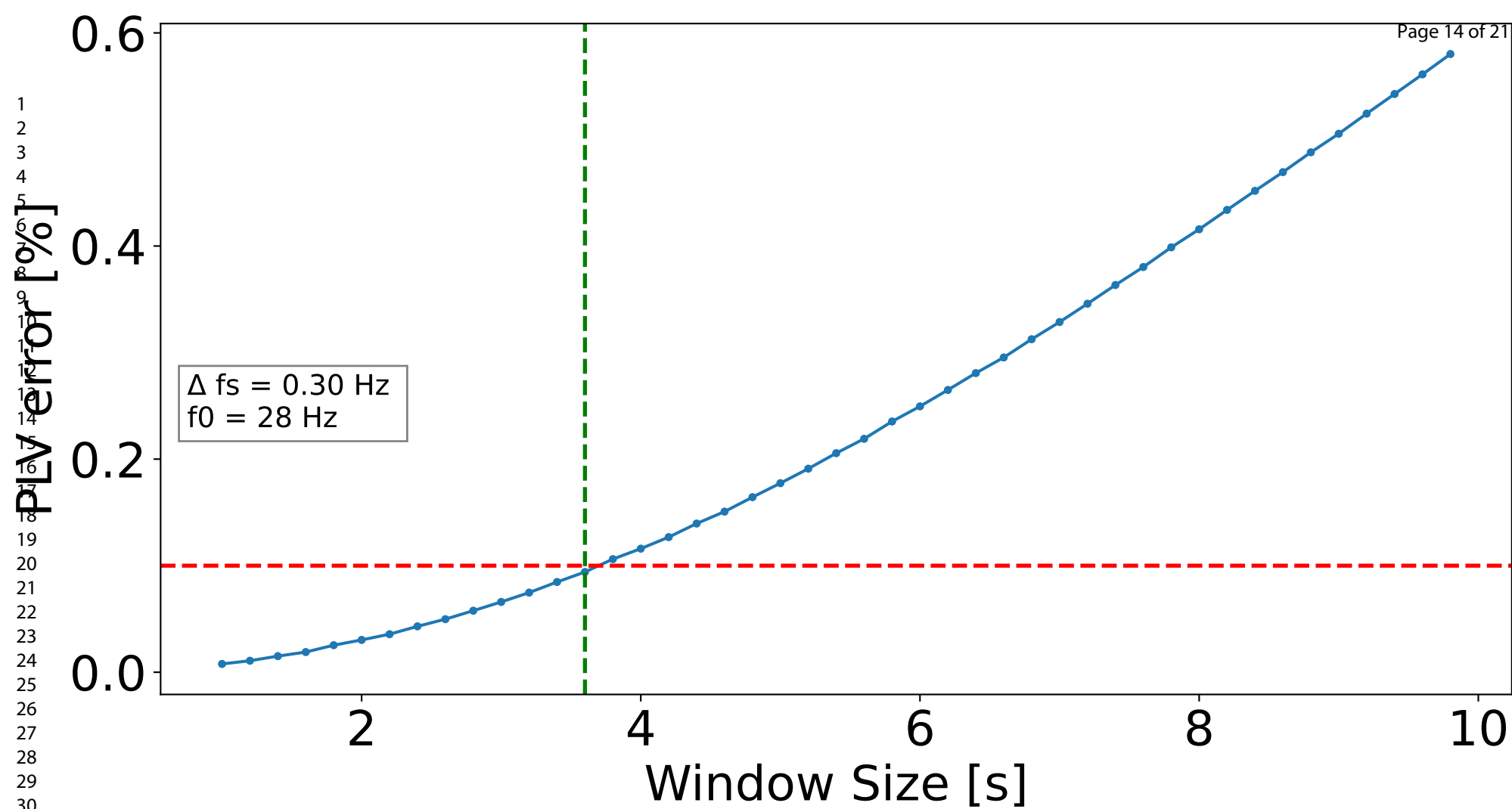
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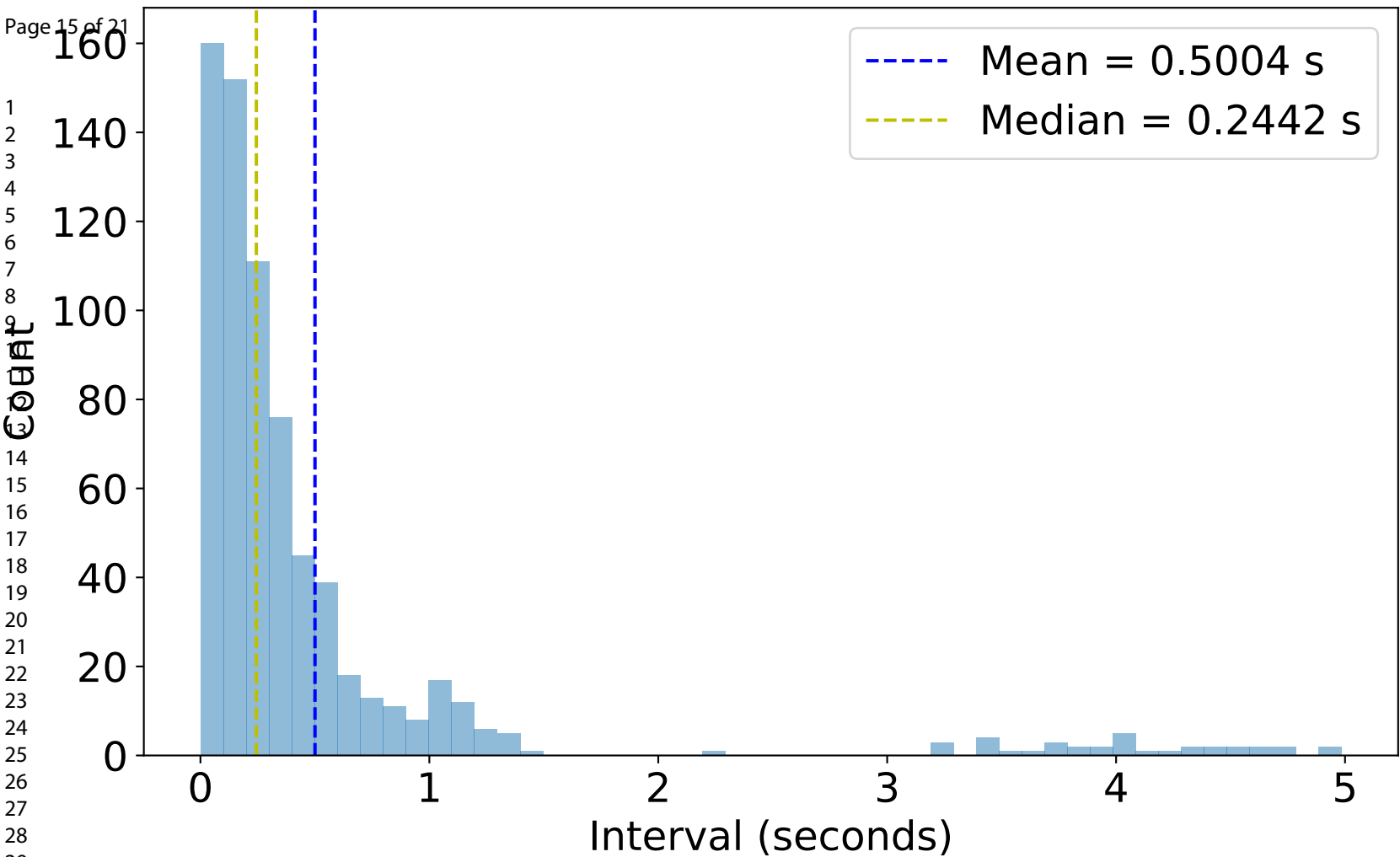






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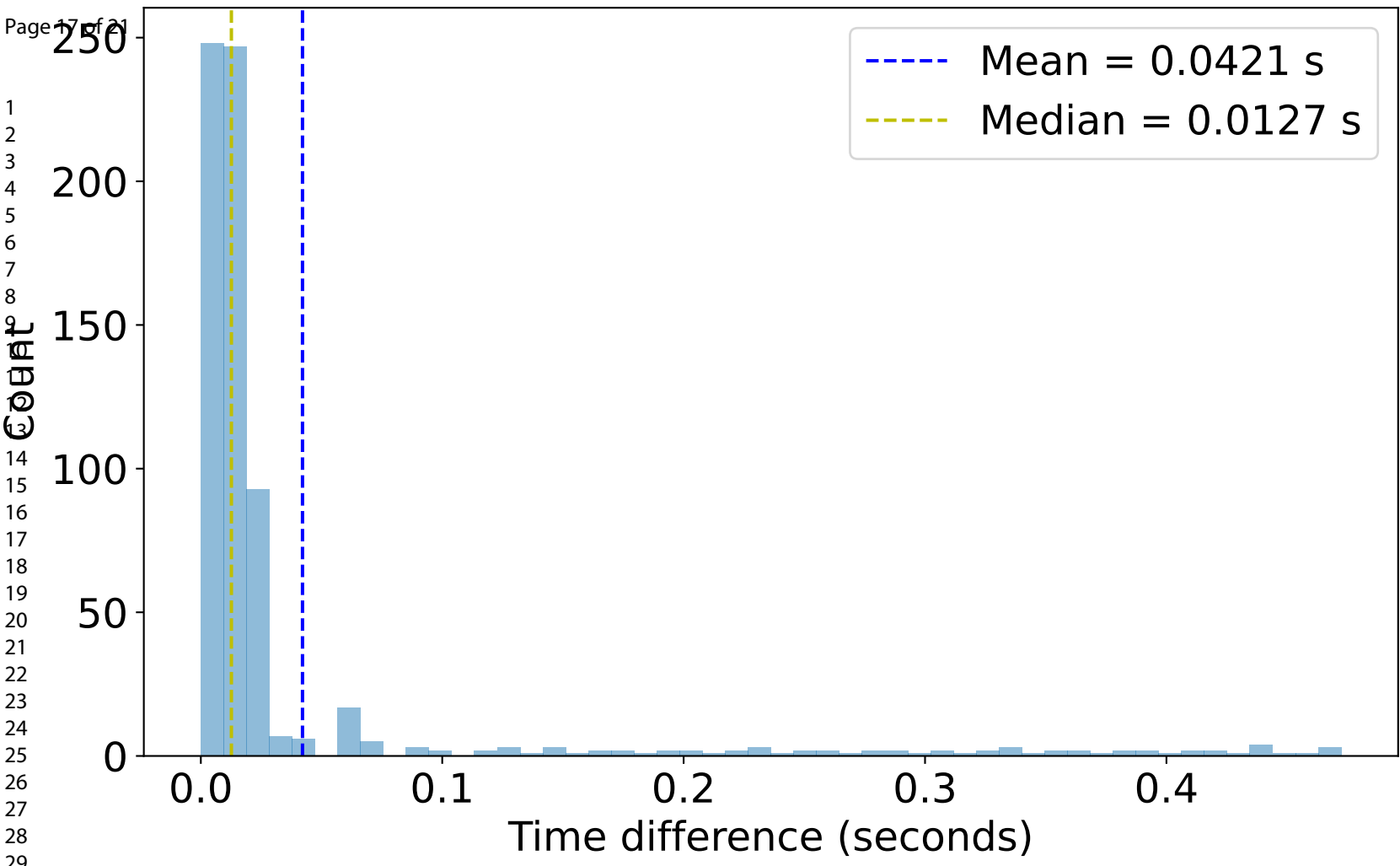




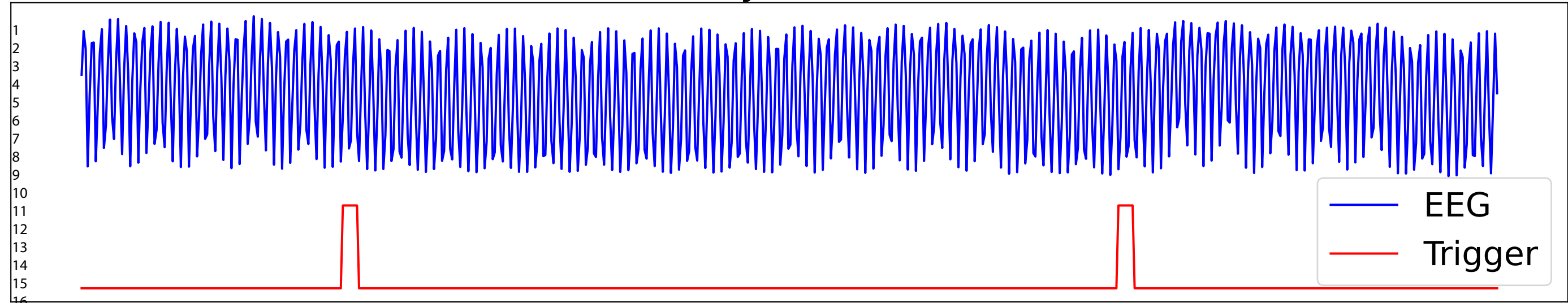
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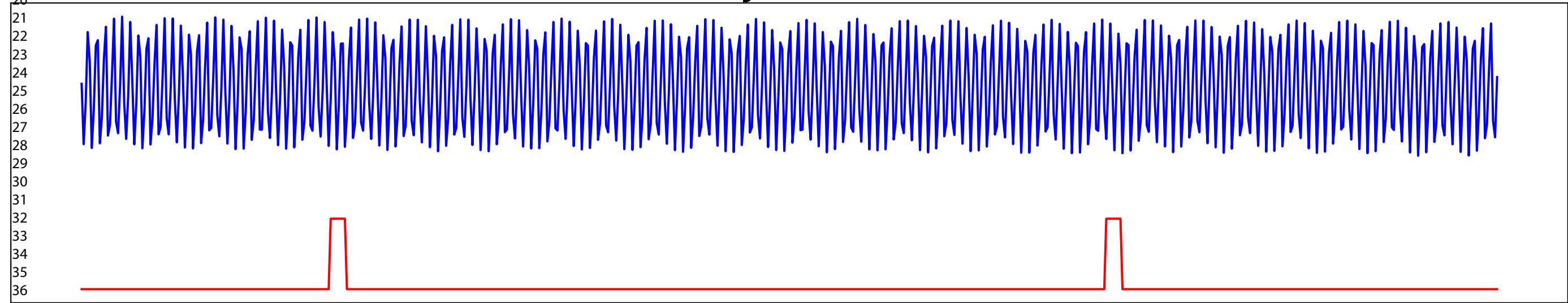
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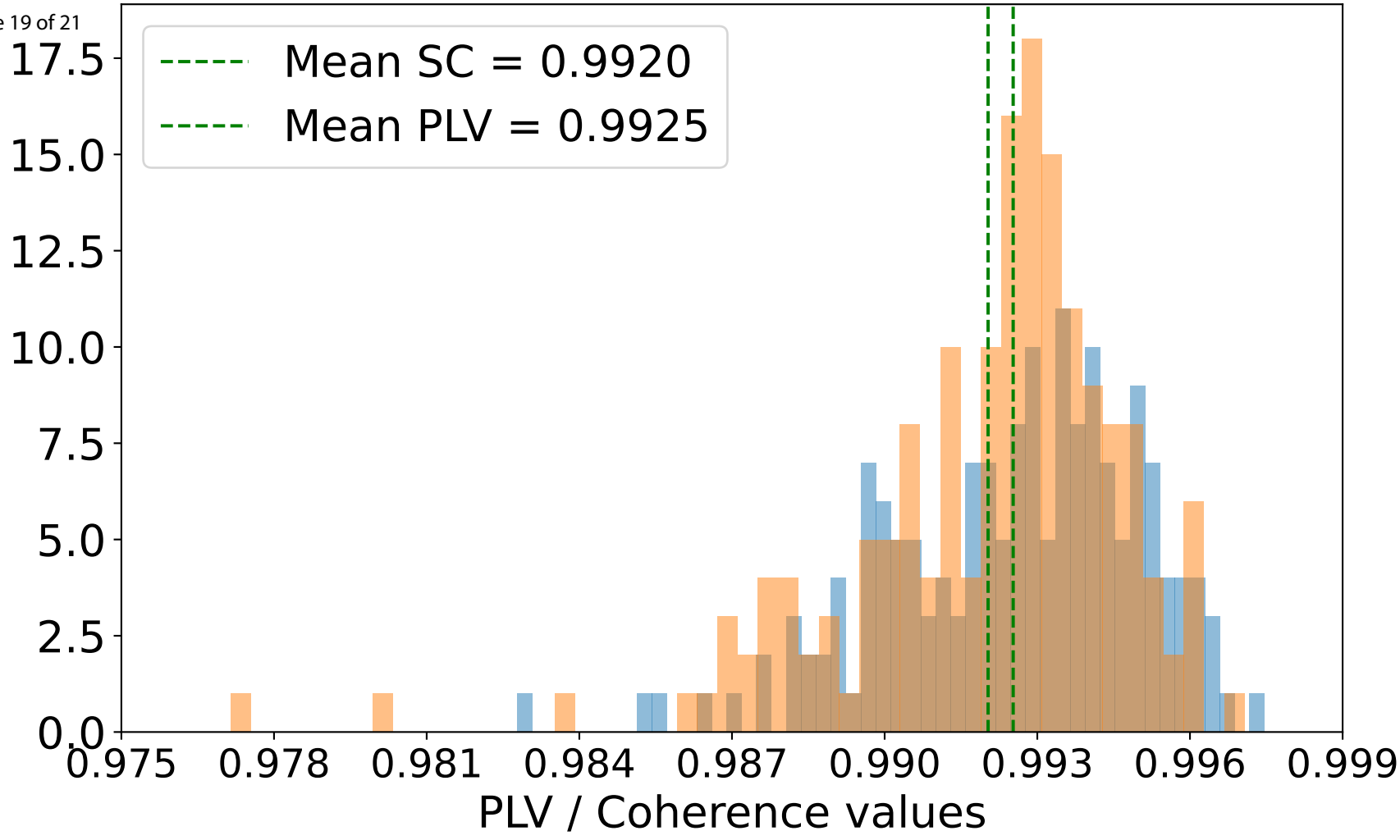
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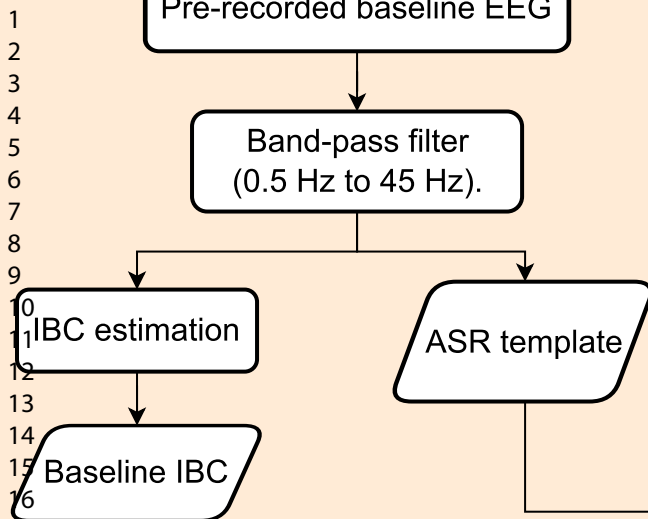
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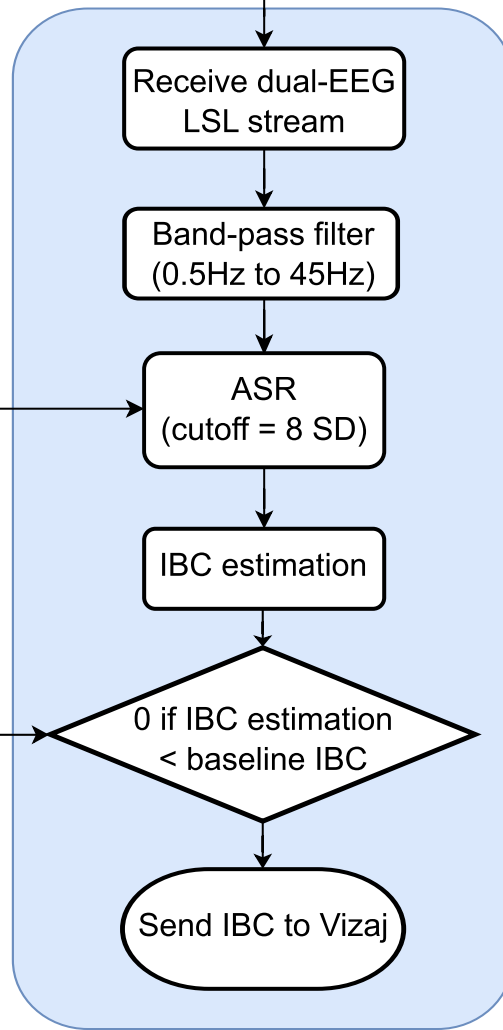
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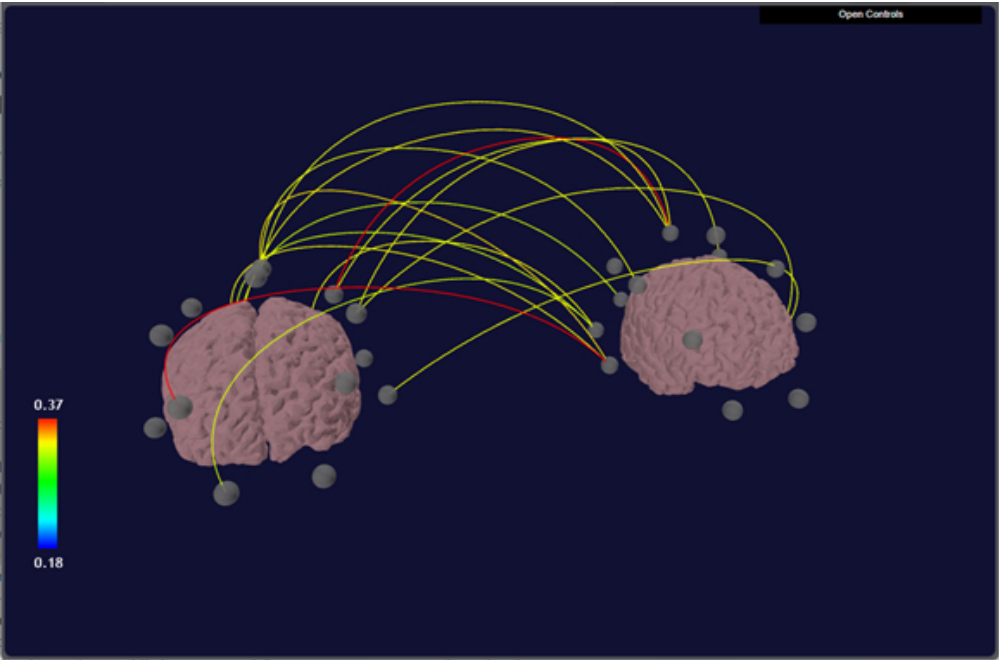
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